ALICE L WALTON SCHOOL OF MEDICINE

Pericardial Effusion and Cardiac Tamponade

Last updated March 8, 2026 · 13 min read · ScholarRx

Adapted by: Diptiman Bose, Phil Riley, Patrick Brooks, Robert Donnell

Editors: Jordan Finn, Russell Johnson

Adapted from: Pericardial Effusion and Cardiac Tamponade

Listen to this Brick

0:00 / 21:58 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the clinical presentations of pericardial effusion and cardiac tamponade
2. Explain the pathophysiology of pericardial effusion and cardiac tamponade.
3. Interpret the ... al effusion and

blood pressure (see pulsus paradoxus, below).

Compare the normal heart and the heart with cardiac tamponade in Figure 2. We can see how pericardial fluid accumulation can compress the cardiac chambers and limit cardiac output.

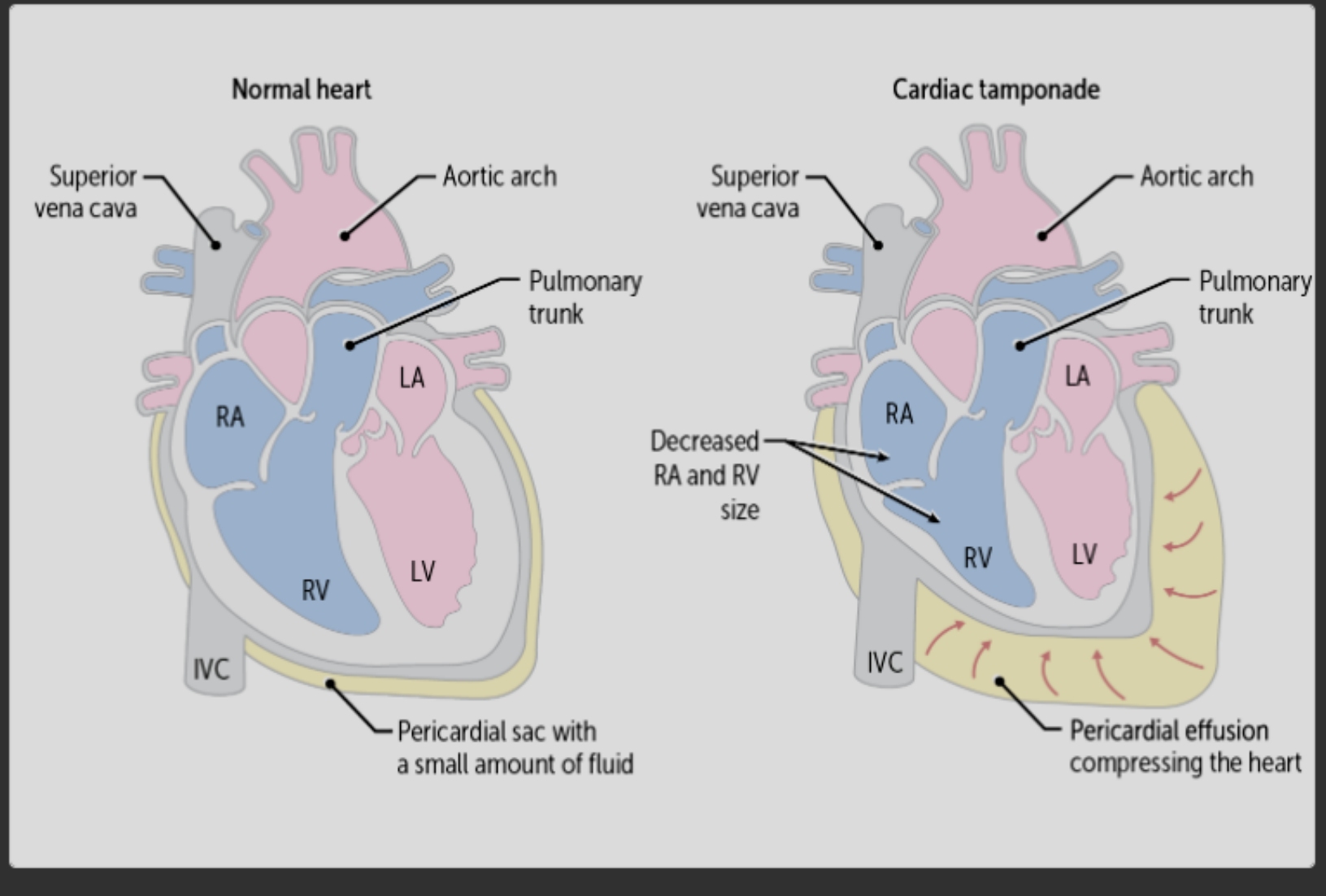

Normal heart
Superior vena cava
Aortic arch
Pulmonary trunk
LA
RA
LV
RV
IVC
Pericardial sac with a small amount of fluid
Cardiac tamponade
Superior vena cava
Aortic arch
Pulmonary trunk
LA
RA
Decreased RA and RV size
RV
LV
IVC
Pericardial effusion compressing the heart

Figure 2. Pathophysiology of cardiac tamponade

The speed of fluid accumulation will determine much of the patient presentation. Slow, chronic fluid accumulation in the pericardium allows for the gradual stretchin... more fluid, making urgent symptoms less common. On the other hand, rapid, acute accumulation

physical exam with the results of electrocardiography (ECG), radiologic studies, and lab tests.

History

Small, chronic pericardial effusions are often discovered incidentally. In contrast, **rapidly developing pericardial effusions are high risk for tamponade** and hemodynamic compromise (eg, dizziness, dyspnea, or syncope due to low cardiac output). Other patient-history findings may help us identify the specific cause that led to the pericardial effusion (eg, malignancy, myocardial infarction, uremia). Chronic effusions can result in tamponade if they reach the limit of pericardial stretching.

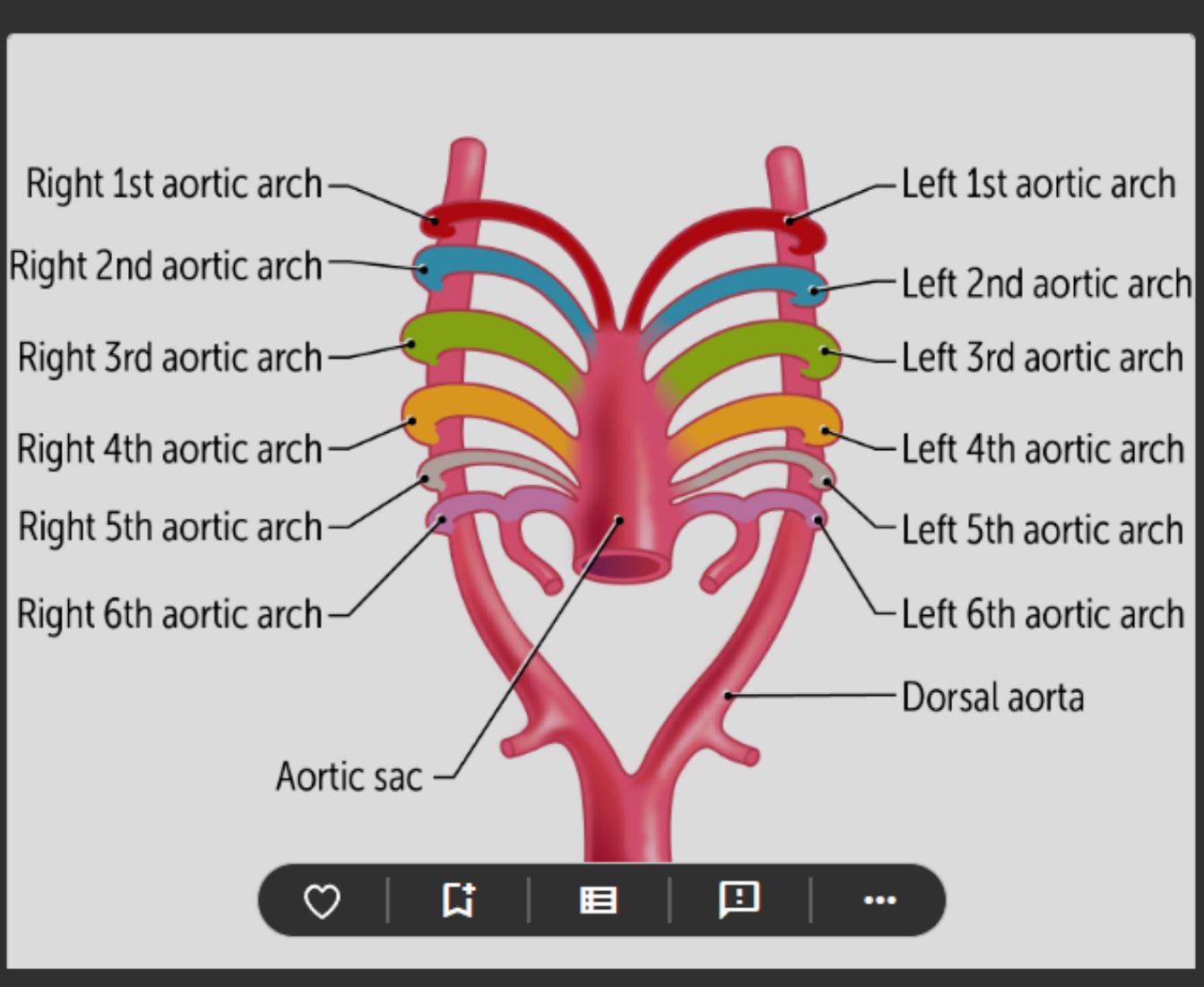
What characteristic of a pericardial effusion is most important regarding the likelihood of developing symptoms?

Physical Examinati

A detailed discussion of the pharyngeal (branchial) arches can be found elsewhere. In this Brick, we focus specifically on how the paired arterial arches develop, remodel, and give rise to the asymmetric great vessels of the adult.

How Do the Aortic Arches Develop? (LO#1)

The **aortic arches** arise as paired arteries that branch from the **aortic sac**, the distal continuation of the truncus arteriosus at the outflow of the embryonic heart. Each arch passes dorsally through its corresponding pharyngeal arch to connect with the **paired dorsal aortae** (Figure 3).


Right 1st aortic arch
Right 2nd aortic arch
Right 3rd aortic arch
Right 4th aortic arch
Right 5th aortic arch
Right 6th aortic arch
Aortic sac
Left 1st aortic arch
Left 2nd aortic arch
Left 3rd aortic arch
Left 4th aortic arch
Left 5th aortic arch
Left 6th aortic arch
Dorsal aorta

ALICE L WALTON SCHOOL OF MEDICINE

Development of the Aortic Arches

Last updated February 28, 2026 · 10 min read · ScholarRx

Adapted by: Alexis Gillett

Editors: Russell Johnson, Jordan Finn

Contributor: Nancy Halliday

Reviewers: Diptiman Bose, Helen Ryals, Carrie Elzie

Adapted from: Development of the Aortic Arches

Listen to this Brick

0:00 / 17:17 1x

Learning Objectives (3) Versions

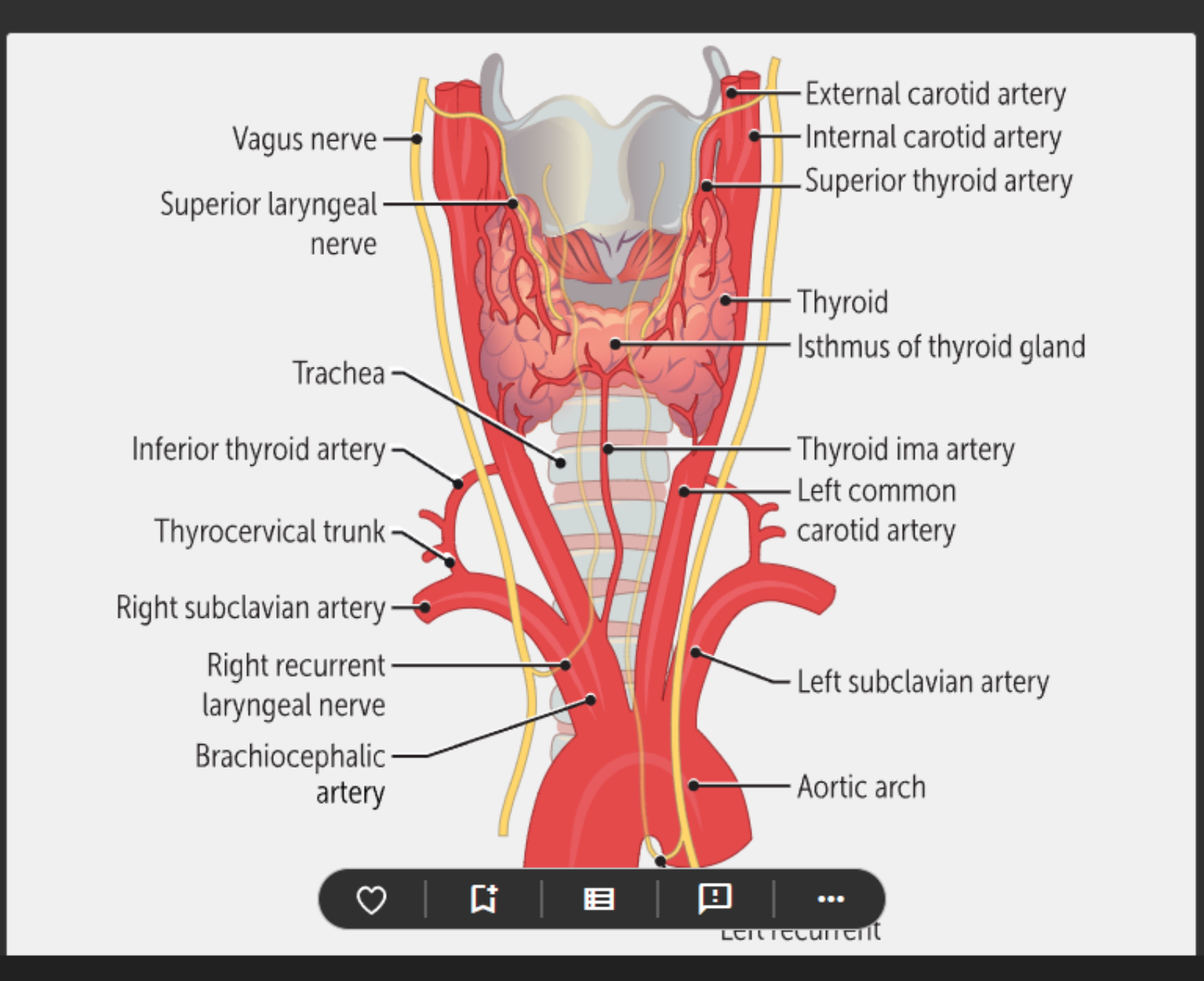
After completing this brick, you will be able to:

1. **Analyze** the sequential development and remodeling of the six embryonic aortic arches to trace how each contributes to specific adult arterial derivatives of the head, neck, and thorax.
2. **Apply** embryologic principles to explain how asymmetric regression of the aortic ... of the right and

arteriosus.

Adult Aortic Anatomy

As shown in Figure 1, the adult aorta gives rise to several major branches whose asymmetric arrangement reflects their embryologic origin. On the **right**, a brachiocephalic trunk divides into the right subclavian and right common carotid arteries. On the **left**, the common carotid and subclavian arteries arise separately from the aortic arch.


Vagus nerve
Superior laryngeal nerve
Trachea
Inferior thyroid artery
Thyrocervical trunk
Right subclavian artery
Right recurrent laryngeal nerve
Brachiocephalic artery
External carotid artery
Internal carotid artery
Superior thyroid artery
Thyroid
Isthmus of thyroid gland
Thyroid ima artery
Left common carotid artery
Left subclavian artery
Aortic arch
Left recurrent

ALICE L WALTON SCHOOL OF MEDICINE

Congenital Heart Defects:

Brick Upgraded

This brick has been upgraded!

While we go to great lengths to preserve your existing work during the upgrade process, you may no longer see certain notes, highlights, or progress on certain learning objectives within the brick. This only impacts brick content that has been modified or removed.

Close

Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Analyze the embryologic development of the heart to identify specific disruptions that lead to congenital anomalies.
2. Create a con... ent, malformation, hemodynamics, and pathophysiology for congenital

What do you want to learn?
ALICE L WALTON SCHOOL OF MEDICINE
Congenital Heart Defects:
Brick Upgraded
This brick has been upgraded!
While we go to great lengths to preserve your existing work during the upgrade process, you may no longer see certain notes, highlights, or progress on certain learning objectives within the brick. This only impacts brick content that has been modified or removed.
Close
Learning Objectives (2) Versions
After completing this brick, you will be able to:
1 Analyze the embryologic development of the heart to identify specific disruptions that lead to congenital anomalies.
2 Create a con... ent, malformation, hemodynamics, and pathophysiology for congenital

What do you want to learn?
ALICE L WALTON SCHOOL OF MEDICINE
Congenital Heart Defects:
Brick Upgraded
This brick has been upgraded!
While we go to great lengths to preserve your existing work during the upgrade process, you may no longer see certain notes, highlights, or progress on certain learning objectives within the brick. This only impacts brick content that has been modified or removed.
Close
Learning Objectives (2) Versions
After completing this brick, you will be able to:
1 Analyze the embryologic development of the heart to identify specific disruptions that lead to congenital anomalies.
2 Create a con... ent, malformation, hemodynamics, and pathophysiology for congenital

What do you want to learn?
ALICE L WALTON SCHOOL OF MEDICINE
Congenital Heart Defects:
Brick Upgraded
This brick has been upgraded!
While we go to great lengths to preserve your existing work during the upgrade process, you may no longer see certain notes, highlights, or progress on certain learning objectives within the brick. This only impacts brick content that has been modified or removed.
Close
Learning Objectives (2) Versions
After completing this brick, you will be able to:
1 Analyze the embryologic development of the heart to identify specific disruptions that lead to congenital anomalies.
2 Create a con... ent, malformation, hemodynamics, and pathophysiology for congenital